

ASTO.P07. Boundary and Freedom: Balance between Control and Chaos

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Context: This document explores the core role of **Boundaries** in **Attribute-Set Transition Ontology (ASTO)**. Audited philosophically, this revision fixes the totalitarian tendency of "positive freedom", introducing **negative freedom**, **right to resist**, and **ethical awakening** mechanisms. **Design of boundaries serves ASTO's ultimate purpose: guarding human homeland before technological singularity, and building better civilization on longer scale.**

Compat Note: Originally ASTOo7.

Declaration: This perturbation may contain cognitive defects, logical holes, and bias. Welcome testing, critique, and overwriting. Retain right to self-mockery and obligation of self-negation.

♥ **Axiom Version Lock**

All axioms cited are based on [ASTO.Po5.Axioms.Phil.v9.1.md](#).

Cognitive Declaration Before Reading

1. Limitations

Interpretation under current cognition and tech conditions, not ultimate truth. ASTO is a description language.

2. Three Meta-Constraints (Three Pre-Action Questions)

Meta-Constraint (Three Pre-Action Questions)	Definition (Summary)	Cognitive Impact
Energy Conservation (Sustainable?)	Under given boundary conditions, continuity of any existence is constrained by energy and dissipation; stability of any structure requires ongoing maintenance cost.	We tend toward effort-saving cognition and model simplification, possibly ignoring complexity and long-chain consequences.
Utility (Can it survive?)	A perturbation that can be retained or expanded must manifest a relatively positive survival benefit (fitness) in its Field.	We tend to focus on parts that form survival feedback (actionable consequences), sacrificing completeness in pure truth-seeking.
Imperfection (Is there a way out?)	There is inevitably a gap between the world and any model; the gap cannot be eliminated, only managed; reserving reversibility and slack for future transition is necessary for continual evolution.	Our cognition is necessarily incomplete; we must presuppose validation/correction/fault-tolerance and reversibility mechanisms.

For detailed definitions, see [ASTO.Po5 Axioms § Meta-Constraints](#).

3. Defect is Source of Creativity

(See [ASTO.Po3.Epistemology § 1.6](#))

4. Don't Treat ASTO as Bible

ASTO is a living tool, not dead dogma. When theory suppresses concrete practical experience, treat it as alienation: revise or discard.

5. Reading Guide

If concepts are unclear, consult:

- [ASTO.P03 Epistemology](#) — Epistemology and Unity of Knowing and Doing
 - [ASTO.P04 Manifesto](#) — Core framework and action program
 - [ASTO.P05 Axioms](#) — Axiom system
 - [ASTO.U02 Glossary](#) — Terminology
-

I. Meaning of Boundary: From Freedom to Structure

1.0 Unity of Knowing and Doing & Boundary

In ASTO epistemology, Unity has three layers: Thinking, Acting, Engineering.
Boundary is the spatial expression of Engineering Layer.

- No boundary = Unexecutable.
- With boundary = Executable.

Boundary defines Freedom; Freedom realizes Unity.

1.1 Interdependence of Freedom and Boundary

Dark Side: When Boundary Becomes Cage

Boundaries can be abused for dystopian systems (Panopticon, Algorithmic Tyranny).

ASTO Immune Mechanism:

1. **Plurality Test:** *If boundary destroys irreplaceability, it is evil.*
2. **Risk Layer Reservation:** *Ethical zone must be guarded by humans.*
3. **Transparency:** *Boundaries must be auditable.*

According to **Axiom 11 (Freedom)**, freedom is ability to change attribute structure *within* constraints. **Theorem 4 (Boundary is Freedom)** states **freedom is not lack of boundary, but certainty of boundary.**

Theorem 4 Derivation (Revised):

1. **Negative Freedom (Bottom Line):** Boundary is **Shield** (Privacy).
2. **Positive Freedom (High Order):** Boundary is **Script** (Actionable area).
3. **Right to Resist:** Freedom includes right to challenge, fork, or exit. System without "exit right" is prison.

In engineering, this is **API Design's Inviolable Zone**.

Code Example:

```
# Wrong example: freedom without boundaries
class UnboundedSystem:
    def execute(self, command):
        return eval(command) # Dangerous! can do anything

# Right example: freedom within boundaries
class BoundedSystem:
    def __init__(self):
        # Oriented Dimension: define inaccessible zones
        self.forbidden_commands = {'rm -rf /', 'shutdown'}
        self.allowed_operations = {'read', 'write', 'compute'}

    def execute(self, command):
        if command in self.forbidden_commands:
            raise SecurityViolation("Inaccessible zone")
        if command.operation not in self.allowed_operations:
            raise PermissionDenied("Operation exceeds boundary")
        return command.execute() # Execute freely within boundary

# Rights Whitebox: explicitly declare user's negative freedoms (default rights)
class RightsWhiteBox:
    def __init__(self, user_id):
        self.user_id = user_id
        # Default rights: no application required, not revocable
        self.default_rights = {
            'data_export',      # right to export data
            'identity_deletion', # right to delete account
            'algorithm_opt_out'  # right to opt out of algorithms
        }

    def get_rights_manifesto(self):
        return {
            "statement": "As a subject, you defaultly own the following protected spaces:",
            "rights": list(self.default_rights)
        }
```

Anti-Pattern Warning: Saying "constraints reduce freedom" is a misunderstanding of Axiom 11. Real freedom is *maximal action space within boundaries*.

1.2 Oriented Dimension: Normative Inaccessibility Structure

⚠️ Oriented Dimension's Po4 Origin

Technical three layers (Norm/Map/Self-Ref) are **Engineering Implementation**, not full connotation.

Ultimate criterion is **Plurality Test**.

Based on **Theorem 4**, Oriented Dimension defines inaccessible areas.

Three Layers:

1. **Norm Layer**: Explicit bans / Revision protocol.
2. **Mapping Layer**: State space forbidden zones.
3. **Self-Ref Layer**: Self-check / Paradox handling.

Fundamental & Taboo:

- **Fundamental (Must Do)**: Min condition (e.g. response < 200ms).
- **Taboo (Cannot Do)**: Red line (e.g. no data deletion).
- Conflict -> **Circuit Break** (Axiom 12).

Risk Layer Protection:

Protect irreducible human attributes (Axiom 7).

Dynamic Governance: List of "Incomputable Zones" must be open to revision.

Engineering Implementation (Complete Risk Layer Protection Pattern):

```
class RiskLayerProtection:
    def __init__(self):
        # Risk Layer: human-reserved incomputable zones
        self.human_reserved = {
            'ethical_judgment',    # e.g. "Should AI be allowed to screen
resumes?"
            'moral_arbitration',  # e.g. "Should we leak user privacy?"
            'privacy_experience'   # e.g. "User data deletion request"
        }
        self.fallback_mode = False # circuit-breaker flag

    def execute(self, action, context):
        # Check whether the action enters a human-reserved zone
        if action.type in self.human_reserved:
            # Force circuit-breaker: stop automation, request human
arbitration
            result = self.request_human_arbitration(action, context)
```

```

        if result == 'REJECT':
            self.fallback_mode = True # trigger graceful degradation
            return self.graceful_degradation(context)
        return result
    else:
        # Algorithmic zone: normal execution
        return action.execute(context)

def request_human_arbitration(self, action, context):
    # Not just routing: must include "ethical awakening"
    # 1) force-display moral consequences (Moral Context) to prevent
    banality of evil
    context_with_impact = self.enrich_moral_context(context)
    # 2) increase operational friction to prevent mindless "Agree"
    # 3) enforce execution delay (duty to think)
    ticket = self.create_high_friction_ticket(action,
context_with_impact)
    return self.wait_for_human_decision(ticket, timeout=24 * 3600,
cooling_off_period=True)

def wait_for_human_decision(self, ticket, timeout,
cooling_off_period):
    # Cooling-off period: even if approved immediately, delay effect
    to prevent impulse
    if cooling_off_period:
        print("Entering ethics cooling-off period: please confirm
final decision after 24 hours.")

        # Dissenter protection: log and protect refusal/non-execution
        cases as evolution samples
        self.log_dissent_case(ticket)

        # ... actual waiting logic ...
        return "PENDING"

def log_dissent_case(self, ticket):
    # Dissent is not error; it is valid feedback to boundaries
    print(f"Logged dissent case: {ticket['action']}. This case enters
the next boundary audit.")

def enrich_moral_context(self, context):
    # Consequence simulation: if approved, what happens?
    return {
        'original': context,
        'impact_simulation': '⚠ Warning: approving this operation
exports 12,000 users\' data into an uncontrolled zone.',
        'accountability': 'This operation will be permanently
recorded in your ethics log.'
    }

```

```

def graceful_degradation(self, context):
    # Degrade after circuit-breaker: return cached data, provide
    limited service, be explicit
    return {
        'status': 'DEGRADED',
        'message': 'This request requires human review; temporarily
returning cached result.',
        'data': self.get_cached_data(context)
    }

def create_high_friction_ticket(self, action, context):
    return {
        'action': action.type,
        'context': context,
        'status': 'PENDING_HUMAN_REVIEW'
    }

```

1.3 Control and Structural Stability

According to **Axiom 1 (Structural Homeostasis)**, any observable system is an anti-noise structure that dynamically maintains a low-entropy state. This is crucial to understanding boundaries.

Encapsulation as Anti-Corruption Layer:

```

# Wrong: no boundary; noise directly invades the core
class NoBoundarySystem:
    def process(self, user_input):
        return eval(user_input) # dangerous: unfiltered noise

# Right: build an anti-noise boundary layer
class BoundarySystem:
    def __init__(self):
        self.validator = InputValidator()
        self.sanitizer = InputSanitizer()

    def process(self, user_input):
        # Boundary layer: shield noise
        if not self.validator.validate(user_input):
            raise ValidationError("Input violates boundary conditions")
        clean_input = self.sanitizer.sanitize(user_input)
        return self._core_process(clean_input)

    def _core_process(self, clean_input):
        # Core layer: process clean data
        return process(clean_input)

```

Anti-Pattern Warning: Treating "openness" as always superior to "boundaries" violates Axiom 1. Without boundaries, there is no system.

1.3.1 Boundary Dilemma: Twitter API Limit Collapse (2023)

Background: Twitter, to control costs, suddenly lowered the free API cap from ~100k tweets/month to ~1k.

Boundary Design Errors:

1. **Norm Layer defect:** no advance notice (violated revision protocol)
2. **Mapping Layer defect:** the forbidden zone (1k) fell far below the ecosystem's Fundamental baseline
3. **Self-Ref Layer defect:** no appeal mechanism (paradox handling missing)

Outcome:

- Thousands of third-party apps collapsed
- Developer community trust collapsed → many apps shut down
- Platform ecosystem value suffered entropy increase

ASTO Diagnosis:

- Fundamental (developer dependency) conflicts with Taboo (cost control)
- Missing transition structure (no gray release / cooling period)
- Did not return to meta-layer (human) arbitration (no community deliberation)
- **Failed Plurality Test:** destroyed the possibility of dialogue

Lesson: Boundary transition is not a technical problem; it is a **governance** problem. Boundaries that fail the Plurality Test are destined to disintegrate.

1.3.2 Concealment vs Transparency of Power

Boundaries are never only technical fences; they are also a **distribution of power**: whoever defines the boundary defines what is "executable / unexecutable", "legal / illegal", "visible / invisible".

If boundaries are hidden deep in code, interfaces, or default configs, people only receive "results of governance", not "participation in governance". Two degradations follow:

- **Negative freedom is eroded:** seemingly no coercion, but you have no refusal path (no exit right).
- **Positive freedom is monopolized:** you cannot participate in how rules are generated and changed (no dissent right).

Therefore, any boundary design should satisfy a minimal **Boundary Contract Theory**:

1. **Explicit**: rules, reasons, responsibility chain, and which Fundamental/Taboo it touches must be visible in docs/metadata.
2. **Contestable**: there must be a **Meta-Protocol** allowing challenges and reviews of the boundary itself (including right to resist / appeal).
3. **Exitable**: if I do not accept the boundary, I must be able to leave with my data intact, or choose an alternative path without harming others.

Anti-Pattern Warning: any boundary that "decides for users" while offering no explanation, no contestation, and no exit is **technological totalitarianism**.

1.3.3 Genealogical Inquiry of Boundary

Boundaries are not neutral engineering artifacts. Each boundary has its **historicity**. Before designing or revising a boundary, ask:

1. **Origin**: When was it set, by whom, and to resolve what concrete interest conflict?
2. **Power**: What historical power relations does it extend? Whose vested interests does it protect? Whose emergence space does it constrain?
3. **Truth effect**: Is it manufacturing an illusion of "this is the only possible reality", thereby killing imagination of other possibilities?

Architecture must pass genealogical audit. If a boundary exists only to prolong an obsolete oppressive structure, it must be dissolved.

1.4 Dynamic Management of Contradiction

Axiom 9 (Internal Tension): contradiction is not an error but a source of tension that maintains and drives evolution.

Three Core Tensions:

1. *Structure vs Agency: boundary stability vs freedom's perturbation*
2. *Order vs Chaos: certainty of control vs creativity of emergence*
3. *Algorithm vs Ethics: efficiency of automation vs human value judgment*

Boundary Jump: when environment changes invalidate an old boundary, the system must perform a boundary jump. It corresponds to the **Seven Orders**:

Seven Orders	Boundary Jump Stage	Engineering Practice
0. Awakening	Realize boundary failure	"This is not a bug; it's a structure problem"
1. Perceive	Detect failure signals	Alerts: error rate > 5% threshold
2. Resolve	Diagnose root conflict	RCA: old rule vs new scenario
3. Intervene	Define new intent	Design review: new boundary spec draft
4. Design	Design transition structure	Dual-write + gray release
5. Materialize	Implement boundary switch	Deploy with feature flags
6. Retrospect	Validate effectiveness	A/B test: compare error rates
7. Dissolve	Retire old boundary	Remove legacy boundary code

Engineering Example (with Seven Orders annotations):

```
class BoundaryJumpSystem:
    def __init__(self):
        self.v1_boundary = LegacyBoundary() # old boundary (order)
        self.v2_boundary = None              # new boundary (to be
designed)
        self.transition_mode = False         # transition flag

    # 1) Perceive: detect boundary failure signals
    def detect_tension(self):
        error_rate = self.calculate_error_rate()
        user_complaints = self.get_complaint_count()

        # 2) Resolve: diagnose root cause
        if error_rate > 0.05 or user_complaints > 100:
            return self.run_root_cause_analysis()
        return None

    # 3) Intervene + 4) Design: define new intent and design transition
structure
    def initiate_boundary_jump(self, new_boundary_spec):
        self.v2_boundary = NewBoundary(new_boundary_spec)
        self.transition_mode = True
        return self.run_dual_write() # dual-write verification
```

```

# 5) Materialize: implement boundary switch
def deploy_new_boundary(self, traffic_percentage=10):
    # Gray release: gradually shift traffic
    self.feature_flag.set_traffic_split(
        v1=100 - traffic_percentage,
        v2=traffic_percentage
    )

# 6) Retrospect: validate new boundary
def validate_new_boundary(self):
    v1_error_rate = self.v1_boundary.get_error_rate()
    v2_error_rate = self.v2_boundary.get_error_rate()
    return v2_error_rate < v1_error_rate # A/B compare

# 7) Dissolve: retire old boundary
def finalize_boundary_jump(self):
    if self.validate_new_boundary():
        self.v1_boundary.deprecate()
        self.v1_boundary = self.v2_boundary
        self.v2_boundary = None
        self.transition_mode = False
        return "Boundary jump complete"
    else:
        return "Rollback to old boundary"

```

Anti-Pattern Warning: refusing necessary boundary jumps out of fear of temporary instability eventually collapses the system.

1.5 Dissolve: Natural End of Old Boundary

In the Seven Orders, the final order is **Dissolve**. Dissolve is not failure; it is the natural end of a cycle.

What Dissolve means:

- The old boundary has fulfilled its historical mission and should be dissolved
- Dissolve frees resources and makes room for new boundaries
- Dissolve is the precondition of a new cycle — without dissolve, there is no renewal

***Political-philosophy note:** Dissolve clears space for a new beginning. As Arendt says, humans have **Natality**. Dissolve is not mere destruction; it is the active ending of an old causal chain so that a real restart becomes possible.*

Dissolve vs Collapse:

- **Collapse** (Six Stages #5): passive, destructive, unplanned structural failure

- **Dissolve** (Seven Orders #7): active, constructive, planned recycling

Engineering practice:

- deprecated marker → dissolve preview
- delete old tables after DB migration → dissolve execution
- sunset API versions → dissolve completion

Boundary defines freedom; Dissolve welcomes renewal.

II. Boundary and Chaos: Temptation of Freedom

2.1 Necessary Structure in Chaos

Axiom 0 (Entropy Increase). Boundaries manage chaos.

2.2 Tension between Motility and Freedom

Freedom (perturbation) disturbs stability. Boundary design must provide space for freedom to guide system to new track without collapse.

2.3 Human vs Machine

Human role: Arbiter and Guide (Axiom 7).

III. Freedom and Control: Final Harmony

3.1 Structuring Freedom

Freedom is realized through clear rules (Axiom 1 + Axiom 12).

Conclusion: Freedom and Boundary are complementary.

Relation with ASTO.P09

P07: How to design boundary?

P09: Why boundary becomes totalitarian tool?

Appendix: Prerequisite Concepts Quick View

If you have not read other ASTO documents, here are the core concepts needed to understand this article:

Attribute-Set

An Attribute-Set is the set of attributes of existence that can be pointed to in a time slice; the transition of Attribute-Sets constitutes the entire history of existence.

Intuitive mapping (for quick entry, not replacing normative definitions): we do not understand the world as fixed "entities (things)"; an "object" can be understood as a temporary aggregation of attributes under specific space-time conditions.

The Five (Five States)

Evolution of spatial forms of existence:

In-itself → Consensus → Encoded → Materialized → Oriented

The Six (Six Stages)

Dynamic process of existence in time:

Chaos → Order → Flux → Pulse → Collapse → Re-origin

The Seven (Seven Orders)

Human intervention action cycle:

Awakening → Perceive → Resolve → Intervene → Design → Materialize → Retrospect → Dissolve

Three Layers of Unity of Knowing and Doing

Layer	"Knowing"	"Doing"	Characteristics
Thinking Layer	Know + Approve	Not yet acted	Will established
Action Layer	Approval driven	Doing, not necessarily right	In trial and error
Engineering Layer	Executable Norm	Automatic execution	Verifiable closed loop

Fundamental & Taboo

- **Fundamental:** minimum conditions the system must satisfy (Must Do)
 - **Taboo:** boundaries the system must never violate (Cannot Do)
-

Appendix C: Ethical Audit Checklist for Boundary Design

Before designing any system boundary, answer:

Plurality Test (Red Lines)

- ☐ Does this boundary turn users into replaceable IDs? (X triggers red line)
- ☐ Does this boundary eliminate the possibility of dissent? (X triggers red line)
- ☐ Does this boundary make behavior fully predictable? (X triggers red line)
- ☐ **Publicity Test:** Does it create/maintain a public space where plural subjects can meet, dialogue, and act? (X if no, treat as atomizing isolation)
- ☐ **Right-to-Resist Test:** Can users exit the system with their data intact? (X if no, treat as prison)
- ☐ **Competitiveness Test:** Is there at least one viable alternative system so the exit right is meaningful? (X if monopoly, exit right is empty talk)

Risk Layer Test (Circuit Break)

- ☐ Is any incomputable zone wrongly automated? (X triggers circuit-breaker)
- ☐ Does human arbitration include **ethical awakening** (consequence display + operational friction)? (✓ required)

Anti-Abuse Test (Security)

- ☐ Are boundary rules transparent and auditable? (✓ required)
- ☐ Is there an appeal and revision mechanism? (✓ required)
- ☐ Have malicious-abuse scenarios been tested? (✓ required)

Decision rule:

- If any Plurality Test triggers a red line → boundary design must be discarded
 - If Risk Layer Test fails → trigger circuit-breaker and return to human arbitration
-



Methodology Origins and References

The ASTO system is distilled from the engineering practice of ODD (Output-Driven Development).

References (Available Offline):

- DOI: 10.5281/zenodo.18207648
- BibTeX:

```
@misc{odd_zenodo_18207648,  
  author    = {Yi Fu},  
  title     = {Output-Driven Development: A Paradigm Shift in AI-Assisted  
Software Engineering},  
  year      = {2026},  
  publisher = {Zenodo},  
  doi       = {10.5281/zenodo.18207648},  
  url       = {https://doi.org/10.5281/zenodo.18207648}  
}
```



Functional Tree

ASTO Documentation System

├─ ☀ P-Series: Philosophy

| ├─ [P01. NotThis] (./ASTO.EN.P01.NotThis.Phil.P.v8.0.md) (Manifesto
of Theoretical Immunity)

| ├─ [P02. Prologue] (./ASTO.EN.P02.Prologue.Phil.v8.0.md) (Negative
Guidance and Path Branching)

| ├─ [P03. Epistemology]
(./ASTO.EN.P03.Epistemology.Phil.Paper.v1.1.md) (Inevitability of
Cognitive Errors)

| ├─ [P04. Manifesto] (./ASTO.EN.P04.Manifesto.Phil.v12.5.md)
(Structural Situation and Action Program)

| ├─ [P05. Axioms] (./ASTO.EN.P05.Axioms.Phil.v9.1.md) (System
Thermodynamics and Structural Ontology)

| ├─ [P06. Values] (./ASTO.EN.P06.Values.Phil.v1.0.md) (Plurality Test
and Ethical Circuit Breaker)

| ├─ [P07. Freedom] (./ASTO.EN.P07.Freedom.Phil.v9.0.md) (Boundary is
Freedom)

| ├─ [P08. Exception] (./ASTO.EN.P08.Exception.Phil.v8.0.md) (Religious
Experience and Interstellar Sovereignty)

| ├─ [P09. Critique] (./ASTO.EN.P09.Critique.Phil.v9.0.md) (Anti-
Totalitarian Charter and System Immunity)

| ├─ [P10. Democracy] (./ASTO.EN.P10.Democracy.Phil.v8.0.md) (Dialogue
Platform and NCP Protocol)

| └─ [P11. Resilience] (./ASTO.EN.P11.Resilience.Phil.v8.0.md) (Self-
Immunity and Antifragility)

```
|  └─ [P12. WhiteSpace] (./ASTO.EN.P12.WhiteSpace.Phil.v1.0.md)
|  (Reserved Extension Space)
|  └─ [P13. Epilogue] (./ASTO.EN.P13.Epilogue.Phil.v8.0.md) (Ultimate
|  Concern of the System)
```

(End)